Network Intrusion Detection System Using Snort

CodeAlpha Task 4 Project Report

Introduction

This project implements a Network Intrusion Detection System using Snort on Windows.

Objective

Monitor network traffic, analyze packets, and generate security alerts.

Tools Used

Snort 2.9.20, Npcap, Windows 11, Command Prompt.

Implementation

Configured Snort, created custom rules, validated configuration, and monitored live network traffic.

Custom Rules

ICMP Ping Detection, HTTPS Traffic Detection, and Possible Port Scan Detection.

Testing Results

Snort processed more than 183,000 packets and generated alerts for HTTPS traffic, ICMP traffic, and possible port scans.

Packet Statistics

TCP: 182970, UDP: 352, ICMP: 6 packets observed during testing.

Conclusion

The Snort-based IDS successfully captured packets, analyzed traffic, and generated alerts, demonstrating effective intrusion detection capabilities.